translation from one formulation of a theory to another, equivalent form, but the *number* of space dimensions can change, too.

More and more, these clues point toward the conclusion that the form of spacetime is an adorning detail that varies from one formulation of a physical theory to the next, rather than being a fundamental element of reality. Much as the number of letters, syllables, and vowels in the word *cat* differ from those in *gato*, its Spanish translation, the form of spacetime—its shape, its size, and even the number of its dimensions—also changes in translation. To any given observer who is using one theory to think about the universe, spacetime may seem real and indispensable. But should that observer change the formulation of the theory he or she uses to an equivalent, translated version, what once seemed real and indispensable necessarily changes, too. Thus, if these ideas are right—and I should emphasize that they have yet to be rigorously proven even though theorists have amassed a great deal of supporting evidence—they strongly challenge the primacy of space and time.

Of all the clues discussed here, I'd pick the holographic principle as the one most likely to play a dominant role in future research. It emerges from a basic feature of black holes—their entropy—the understanding of which, many physicists agree, rests on firm theoretical foundations. Even if the details of our theories should change, we expect that any sensible description of gravity will allow for black holes, and hence the entropy bounds driving this discussion will persist and holography will apply. That string theory naturally incorporates the holographic principle—at least in examples amenable to mathematical analysis—is another strong piece of evidence suggesting the principle's validity. I expect that regardless of where the search for the foundations of space and time may take us, regardless of modifications to string/M-theory that may be waiting for us around the bend, holography will continue to be a guiding concept.

The Constituents of Spacetime

Throughout this book we have periodically alluded to the ultramicroscopic constituents of spacetime, but although we've given